6.3 Properties of Radicals

Algebra I 6.3 · Student sheet

Name _____ Period ____ Date _____

I can…

- use the properties of radicals to simplify expressions.
- rationalize the denominator of a radical expression.
- add, subtract, and multiply radical expressions.

Vocabulary · fill in the missing word

Product Property of Radicals — $\sqrt[n]{ab} =$ _____

Quotient Property of Radicals — $\sqrt[n]{a/b} =$ _____

Simplest form — no perfect nth-power factors, no fractions in the radicand, no radicals in a _____

Rationalizing the denominator — _____ radicals from a denominator

Conjugates — _____

Like radicals — same _____ and same radicand

Part 1 · True or false?

Use a calculator to test each statement. Mark it true or false.

Statement	Left side	Right side	True or false?
$\sqrt{4} \cdot \sqrt{9} = \sqrt{36}$			
$\sqrt{100} / \sqrt{4} = \sqrt{25}$			
$\sqrt{9} + \sqrt{16} = \sqrt{25}$			
$\sqrt{50} = 5\sqrt{2}$			
$1/\sqrt{2} = \sqrt{2}/2$			

Two rules that are always true:

Part 2 · Notes: fill in the blanks

Product Property: $\sqrt[n]{ab} =$ _____ **Quotient Property:** $\sqrt[n]{a/b} =$ _____

A radical is in simplest form when there are no perfect nth-power _____, no _____ in the radicand, and no radical in the _____.

To rationalize $5/\sqrt{3}$, multiply by _____.

The conjugate of $\sqrt{3} + 1$ is _____.

Like radicals have the same _____ and the same _____.

Part 3 · Examples (work them with the class)

Example 1

Simplify. Assume all variables are positive.

a. $\sqrt{108}$ b. $\sqrt{9x^3}$

Example 2

Simplify. Assume all variables are positive.

a. $\sqrt{\frac{15}{64}}$ b. $\sqrt{\frac{81}{x^2}}$

Example 3

Simplify.

a. $\sqrt[3]{(-128)}$ b. $\sqrt[3]{(125x^7)}$

Example 4

Simplify $\frac{5}{\sqrt{3}}$.

Example 5

Simplify $\frac{2}{\sqrt{3}+1}$.

Example 6

Simplify $3\sqrt{2}+\sqrt{8}$.

Part 4 · Partner practice

Partner 1

Simplify $\sqrt{72}$.

Partner 2

Simplify $\dfrac{4}{\sqrt{6}}$.